Sales Data Analysis Report

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1 Executive Summary

This report presents a comprehensive analysis of sales transaction data covering **January 2023 to March 2024**. The dataset contains **9,140 transactions** from **300 unique customers** across **10 Indian cities**, spanning **10 product categories** and **105 products**.

Key Highlights:

- Total Revenue generated: **Rs 22.9 Crore**
- Average spend per transaction: **Rs 25,057**
- Top performing category: **Electronics (68% of revenue)**
- Most valuable customer segment: **VIP (54% of revenue)**
- All **12 data quality checks passed** with zero issues
- **300 customer RFM profiles** built with next purchase predictions

2 Dataset Overview

2.1 Raw Data Summary

Metric	Value
Total Rows	9,140
Total Columns	15
File Size (KB)	1196.63
Unique Customers	300
Unique Products	105
Unique Categories	10
Cities Covered	10
Date Range	Jan 2023 to Mar 2024

Table 1: Dataset Overview

2.2 Column Information

Column	Type	Unique Values	Missing
customer_id	character	300	0
customer_name	character	300	0
age	integer	48	0
gender	character	3	0
mobile_number	numeric	300	0
city	character	10	0
purchase_date	character	6285	0
product_category	character	10	0
product_name	character	105	0
quantity	integer	10	0
unit_price	numeric	9033	0
total_spend	numeric	9071	0
payment_method	character	6	0
channel	character	4	0
customer_segment	character	4	0

Table 2: Column Names and Data Types

2.3 Categorical Value Reference

Column	Unique Values
gender	Female, Male, Other
customer segment	VIP, Premium, Regular, New Customer
payment method	UPI, Credit Card, Debit Card, Cash, Wallet, Net Banking
channel	Online, Mobile App, In-Store, Phone
product category	Electronics, Groceries, Clothing, Books, Furniture, Beauty, Automotive, Sports, Toys, Other

Table 3: Categorical Column Values

3 Statistical Summary

3.1 Numeric Column Statistics

Column	Min	Max	Mean	Median	Std Dev	Skewness
age	18.0	65.0	43.1	44.0	13.9	-0.14
quantity	1.0	10.0	2.3	2.0	1.9	1.96
unit_price	41.4	201009.1	10634.3	1183.7	25074.9	3.42
total_spend	41.4	1328474.6	25057.0	2509.4	81166.2	7.39

Table 4: Descriptive Statistics

3.2 Skewness Interpretation

	Column	Skewness	Interpretation
age	age	-0.14	Symmetric
quantity	quantity	1.96	Highly Skewed - Log Transform Applied
unit_price	unit_price	3.42	Highly Skewed - Log Transform Applied
total_spend	total_spend	7.39	Highly Skewed - Log Transform Applied

Table 5: Skewness Analysis

3.3 Categorical Distributions

Value	Count	Percent
Female	3731	40.8%
Male	2809	30.7%
Other	2600	28.4%

Table 6: Gender Distribution

Value	Count	Percent
VIP	4524	49.5%
Premium	2683	29.4%
Regular	1482	16.2%
New Customer	451	4.9%

Table 7: Customer Segment Distribution

Value	Count	Percent
Groceries	1675	18.3%
Electronics	1306	14.3%
Clothing	1039	11.4%
Books	905	9.9%
Furniture	874	9.6%
Beauty	845	9.2%
Automotive	743	8.1%
Sports	636	7%
Other	596	6.5%
Toys	521	5.7%

Table 8: Product Category Distribution

Value	Count	Percent
UPI	2464	27%
Credit Card	2347	25.7%
Debit Card	1813	19.8%
Cash	932	10.2%
Wallet	846	9.3%
Net Banking	738	8.1%

Table 9: Payment Method Distribution

Value	Count	Percent
Online	3361	36.8%
Mobile App	2349	25.7%
In-Store	2278	24.9%
Phone	1152	12.6%

Table 10: Channel Distribution

4 Data Quality Report

4.1 Missing Values

Column	Missing	Pct %	Status
customer_id	0	0	CLEAN
customer_name	0	0	CLEAN
age	0	0	CLEAN
gender	0	0	CLEAN
mobile_number	0	0	CLEAN
city	0	0	CLEAN
purchase_date	0	0	CLEAN
product_category	0	0	CLEAN
product_name	0	0	CLEAN
quantity	0	0	CLEAN
unit_price	0	0	CLEAN
total_spend	0	0	CLEAN
payment_method	0	0	CLEAN
channel	0	0	CLEAN
customer_segment	0	0	CLEAN

Table 11: Missing Value Analysis

4.2 Validation Checks Summary

Check	Issues Found	Result
Missing Values	0	PASS
Full Duplicates	0	PASS
Partial Duplicates (kept)	176	INFO
Age Range 18-100	0	PASS
Quantity > 0	0	PASS
Unit Price > 0	0	PASS
Total Spend > 0	0	PASS
Future Dates	0	PASS
Pre-2020 Dates	0	PASS
Mobile 10 digits	0	PASS
Spend = Qty x Price	0	PASS
Category Values	0	PASS

Table 12: Validation Checks (12 of 12 Passed)

4.3 Outlier Detection

	Column	Lower Fence	Upper Fence	Count	Pct	Action
25%	age	-2	90	0	0%	None
25%1	quantity	-2	6	453	5%	Winsorized 1-99%
25%2	unit_price	-5902	11223	1447	15.8%	Winsorized 1-99%
25%3	total_spend	-12428	23033	1535	16.8%	Winsorized 1-99%

Table 13: Outlier Detection - IQR Method

5 Data Cleaning & Feature Engineering

5.1 Cleaning Steps Applied

Step	Action
1. Column Names	clean_names() - snake_case applied
2. Data Types	age and qty to integer, prices to numeric
3. Mobile Number	Character - 10-digit preserved
4. Purchase Date	ymd_hms() to POSIXct
5. Factors	gender, city, segment, payment, channel, category
6. Duplicates	0 rows removed
7. Recalculate Spend	qty x unit_price rounded to 2 decimal
8. Winsorize Prices	unit_price and total_spend capped 1-99 pct

Table 14: Cleaning Steps Summary

5.2 Feature Engineering - 15 New Columns

Feature	Description
year	Year of transaction
month_num	Month number 1-12
month_name	Month abbreviation
day	Day of month
hour	Hour of transaction
day_of_week	Day of week Sun-Sat
quarter	Q1 to Q4
is_weekend	TRUE if Saturday or Sunday
is_month_end	TRUE if day >= 25
season	Winter/Summer/Monsoon/Autumn
time_of_day	Morning/Afternoon/Evening
age_group	5 age buckets 18-25 to 55+
spend_tier	Low/Mid/High spender
log_unit_price	log1p of unit_price
log_total_spend	log1p of total_spend

Table 15: New Features Created

6 Customer Profiling & RFM Analysis

6.1 RFM Segment Distribution

Segment	Customers	Percent
Loyal Customers	92	30.7%
Lost	85	28.3%
Champions	63	21%
Potential Loyalists	58	19.3%
New Customers	2	0.7%

Table 16: RFM Segment Distribution

6.2 Top 10 Customers by Revenue

ID	Name	City	Transactions	Revenue	RFM
CID188	Rupali Iyer	Bangalore	62	Rs 6,232,431	Champions
CID118	Vaishali Rajan	Hyderabad	65	Rs 5,206,408	Champions
CID201	Sahil Oswal	Mumbai	53	Rs 4,728,082	Loyal Customers
CID051	Ramesh Nair	Hyderabad	63	Rs 4,341,383	Champions
CID231	Parth Menon	Jaipur	61	Rs 4,293,976	Champions
CID032	Meena Venkat	Delhi	61	Rs 3,580,158	Champions
CID241	Umesh Singhania	Kolkata	61	Rs 3,542,978	Champions
CID053	Aisha Roy	Kolkata	67	Rs 3,511,901	Champions
CID002	Sachin Baig	Mumbai	59	Rs 3,322,860	Champions
CID054	Balaji Menon	Jaipur	61	Rs 2,821,577	Champions

Table 17: Top 10 Customers by Revenue

6.3 Frequency Labels

Label	Count	Percent
Bi-Weekly Buyer	131	43.7%
Weekly Buyer	81	27%
Monthly Buyer	51	17%
Occasional Buyer	36	12%
Rare Buyer	1	0.3%

Table 18: Purchase Frequency Labels

6.4 RFM vs Customer Segment

Table 19: RFM vs Customer Segment Cross-Table

	Champions	Lost	Loyal Customers	New Customers	Potential Loyalists
New Customer	0	56	3	2	7
Premium	21	4	31	0	23
Regular	5	25	23	0	21
VIP	37	0	35	0	7

7 EDA Visualizations

7.1 Monthly Revenue Trend

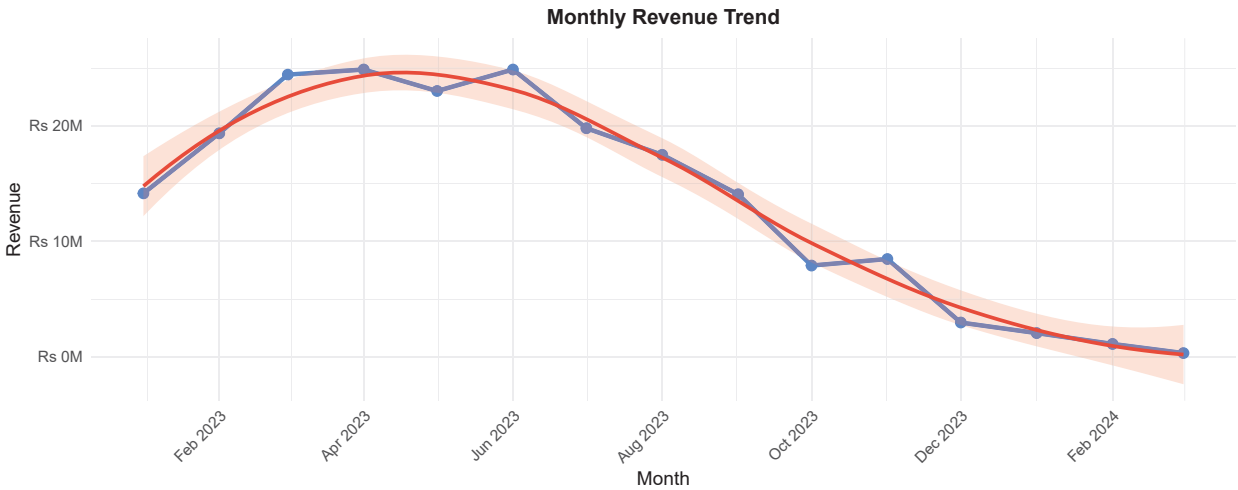

Figure 1: Monthly Revenue - Jan 2023 to Mar 2024

7.2 Revenue by Day of Week & Quarter

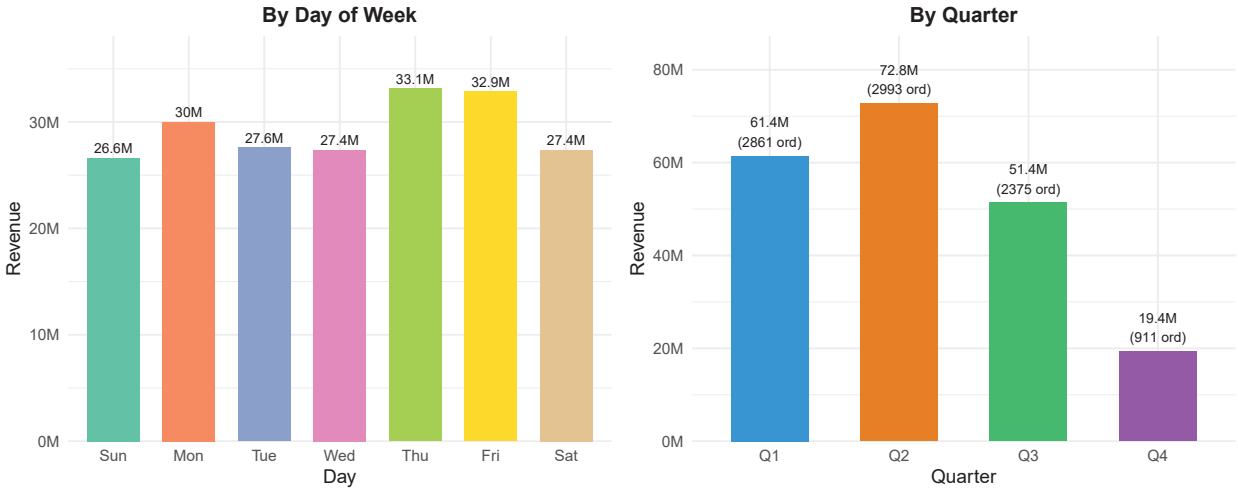

Figure 2: Revenue by Day of Week (left) and Quarter (right)

7.3 Revenue by Season & Category

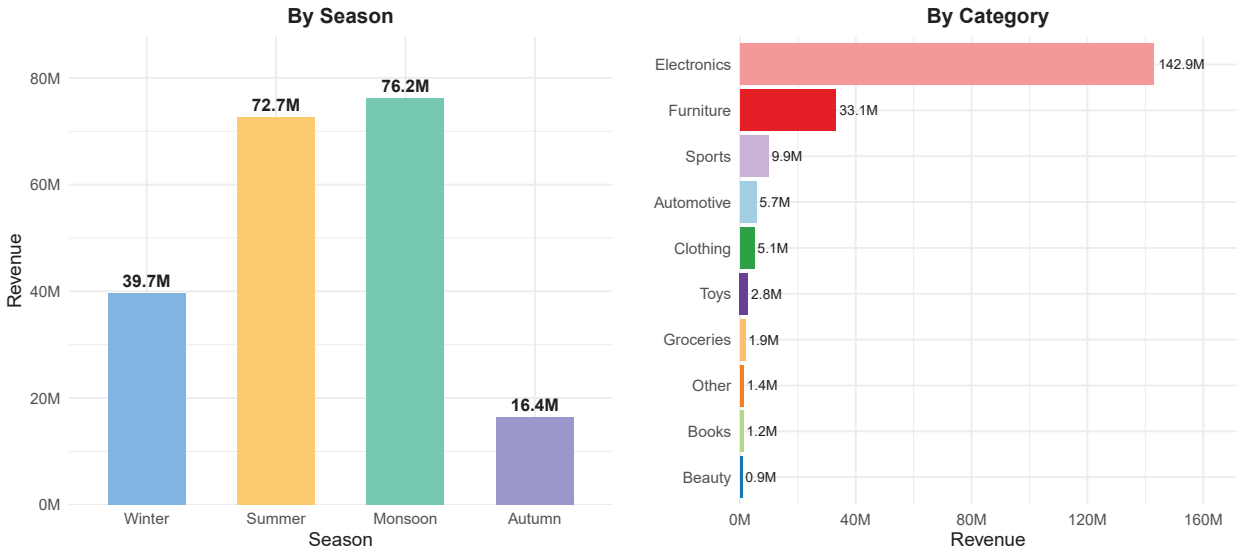

Figure 3: Revenue by Season (left) and Category (right)

7.4 Customer Segment Analysis

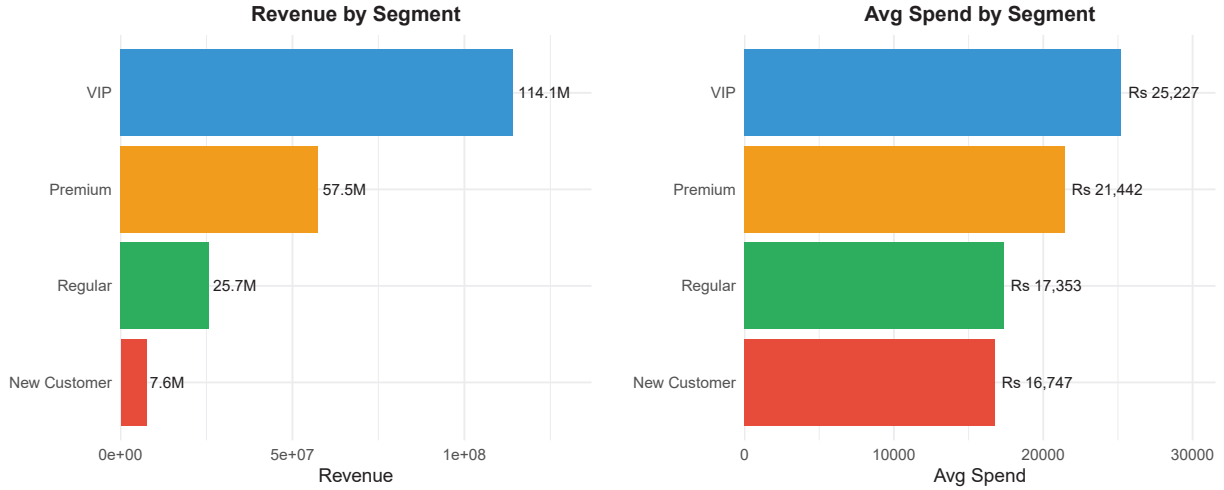

Figure 4: Revenue and Avg Spend by Customer Segment

7.5 Gender & Age Group

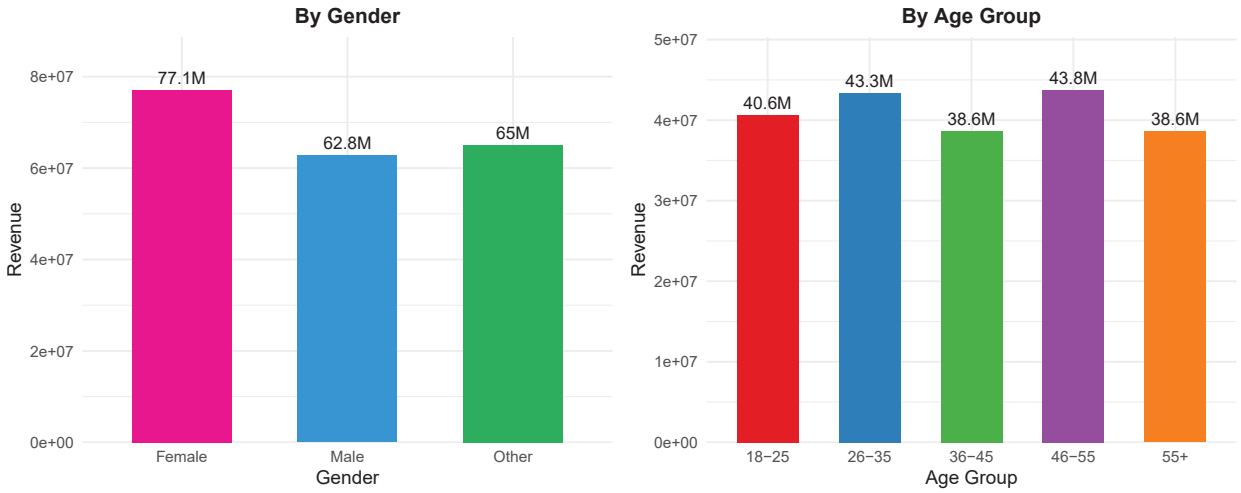

Figure 5: Revenue by Gender (left) and Age Group (right)

7.6 Payment Method & Channel

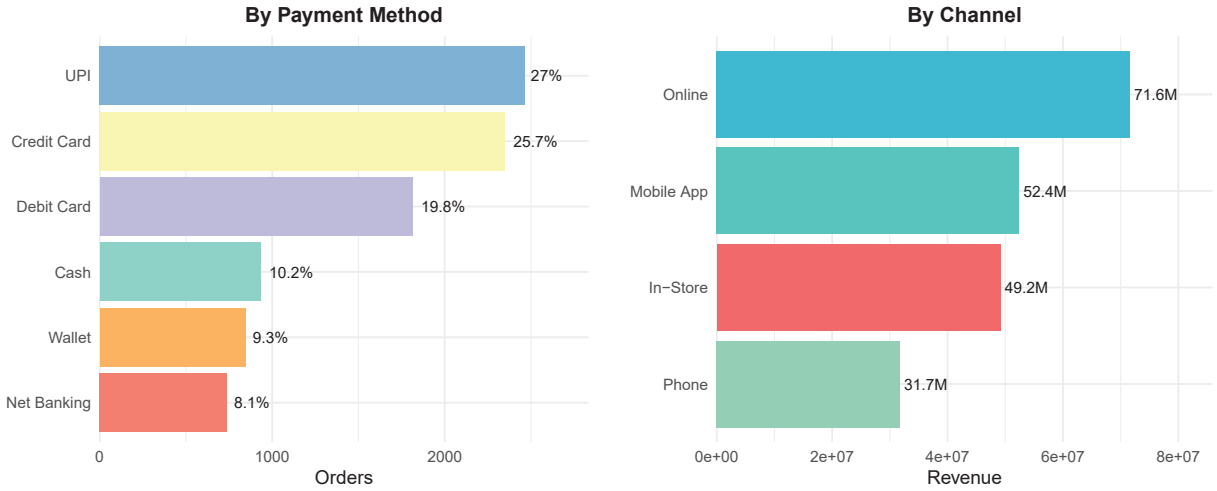

Figure 6: Orders by Payment Method (left) and Revenue by Channel (right)

7.7 City Revenue

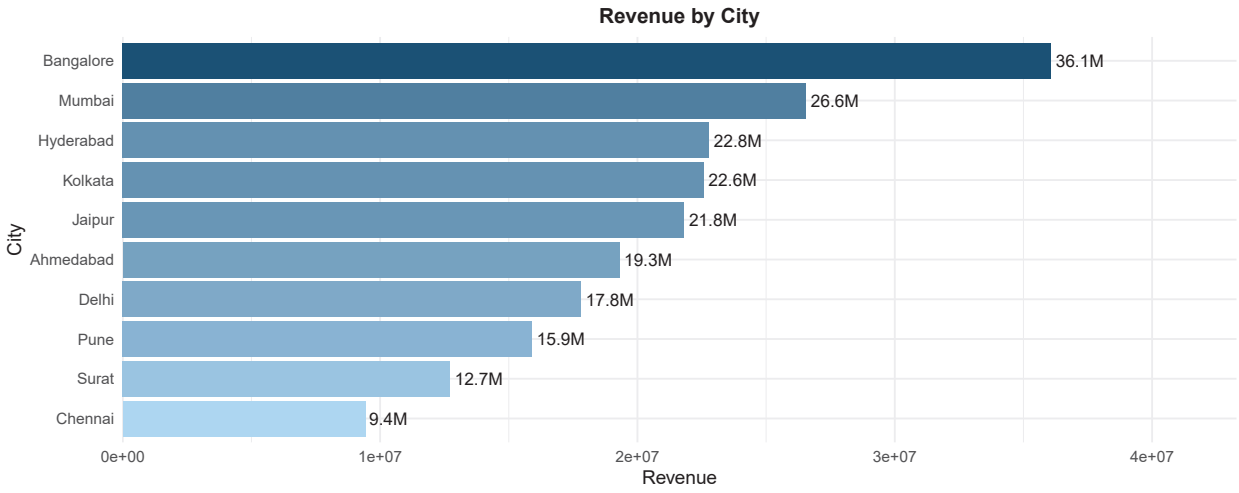

Figure 7: Total Revenue by City

7.8 RFM Segment Distribution

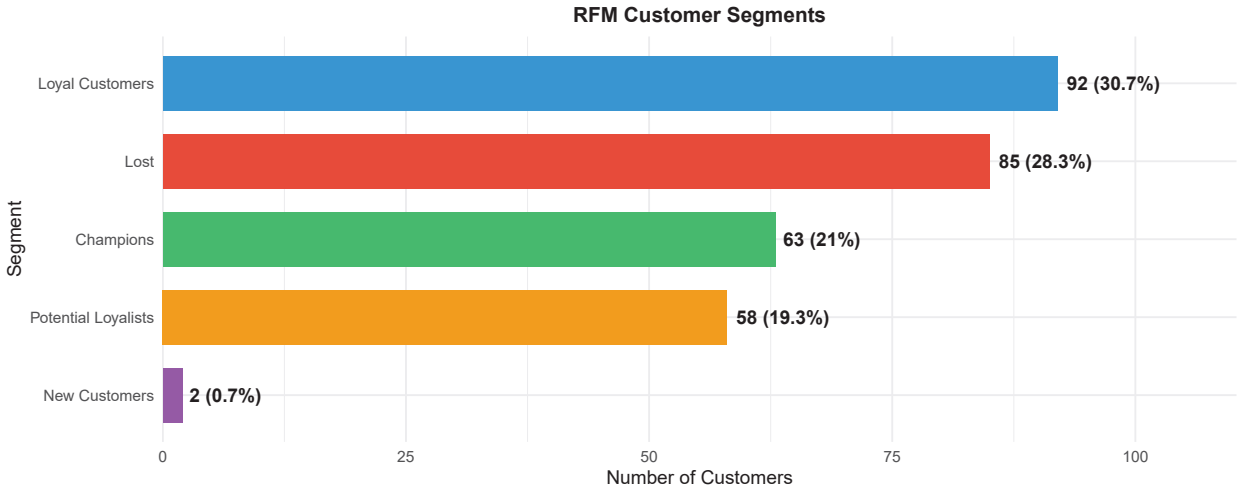

Figure 8: RFM Customer Segment Distribution

7.9 Spend Distribution & Ridge Plot

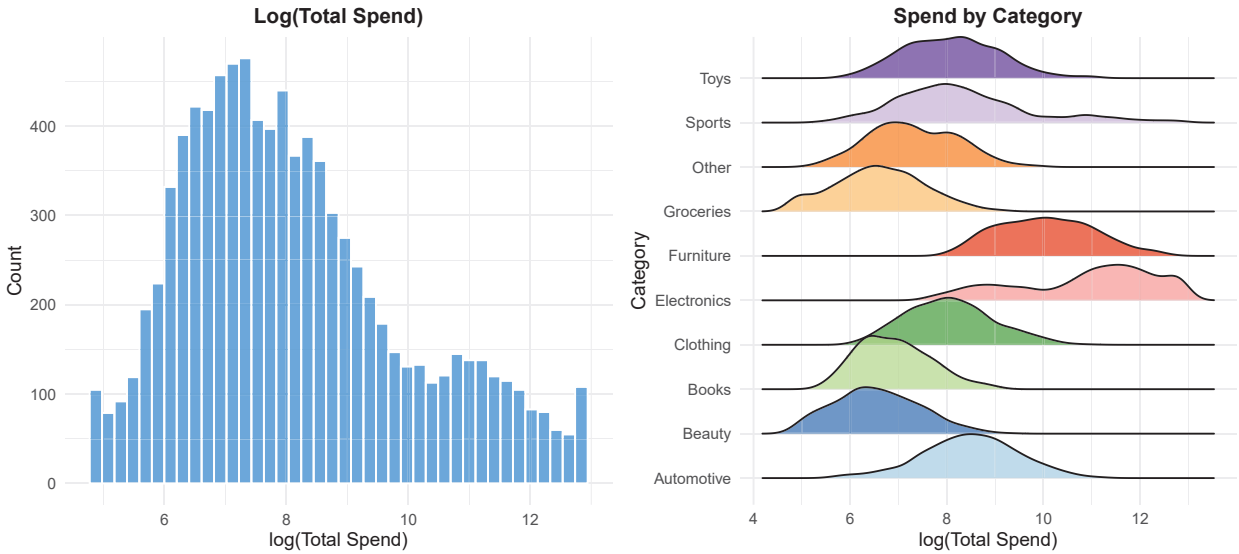

Figure 9: Log Spend Distribution (left) and Ridge by Category (right)

7.10 Correlation Matrix

Figure 10: Correlation Matrix - Numeric Variables

7.11 Top 10 Products by Revenue

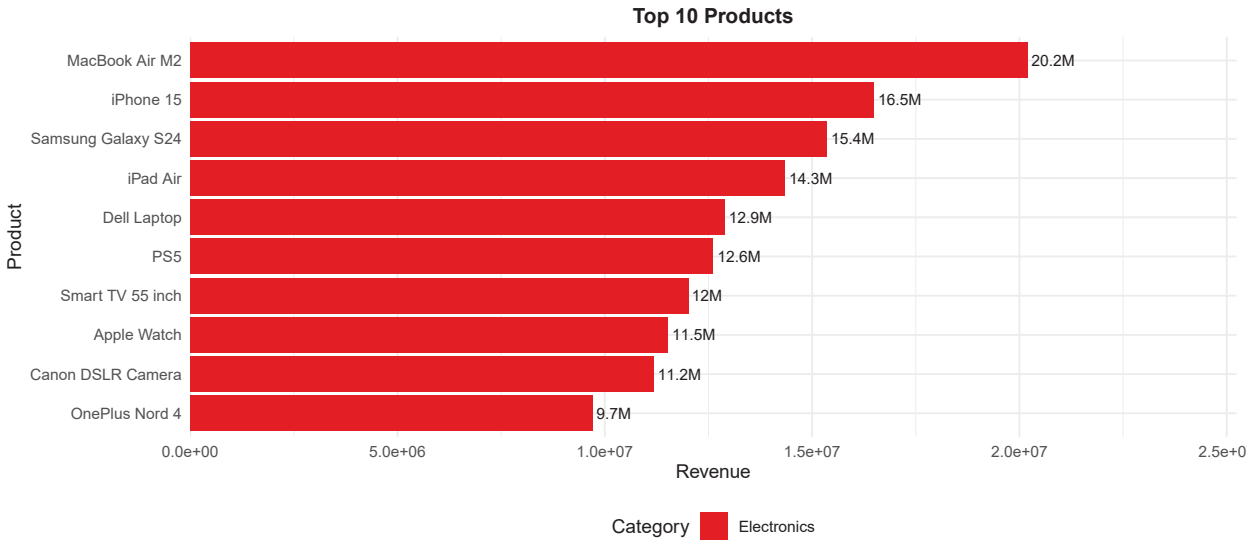

Figure 11: Top 10 Products by Revenue

7.12 Time of Day vs Channel & Category-Payment Heatmap

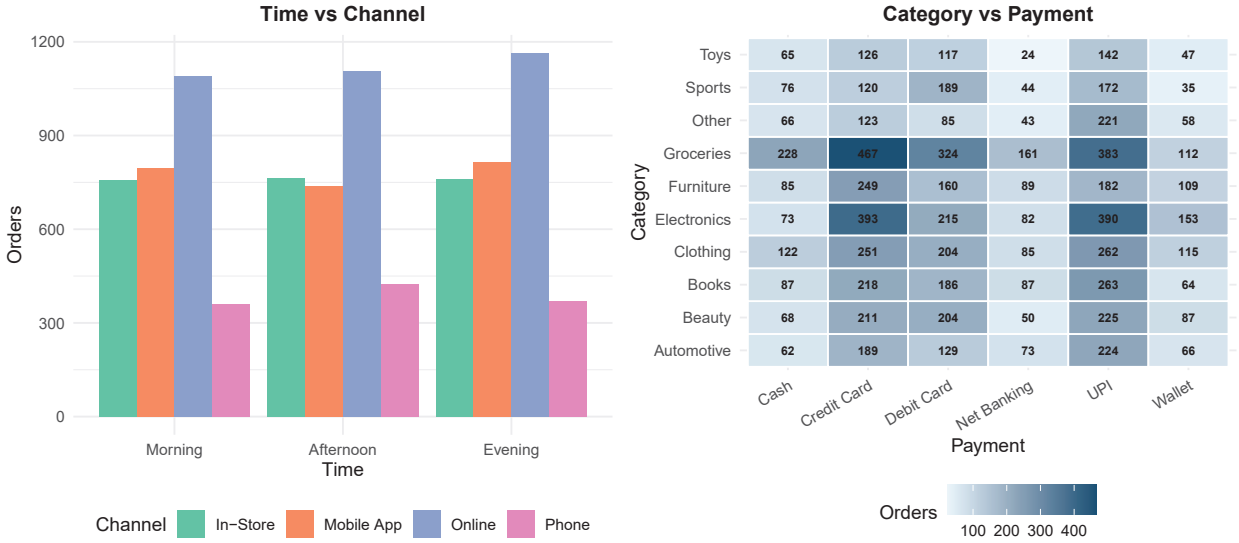

Figure 12: Time vs Channel (left) and Category-Payment Heatmap (right)

8 Output Files

File	Location	Contents
clean_sales_data.csv	Output/Data/	9140 rows x 30 columns
customer_profile.csv	Output/Data/	300 customers x 36 columns
monthly_trend.csv	Output/Data/	15 months aggregated
category_summary.csv	Output/Data/	10 categories aggregated
Sales_Analysis_Report.pdf	Project Root/	This full PDF report

Table 20: All Output Files Generated

9 Key Business Insights

Area	Insight	Priority
Revenue	Total Rs 22.9 Crore in 15 months	INFO
Revenue	Avg transaction value Rs 25,057	INFO
Category	Electronics = 68 pct of all revenue	HIGH
Category	All Top 10 products are Electronics	HIGH
Customer	VIP = 49.5 pct customers, 54 pct revenue	HIGH
Customer	85 Lost customers need re-engagement	HIGH
Channel	Online leads with 36.8 pct of orders	MED
Payment	UPI most preferred at 27 pct	MED
Geography	Bangalore top city at 15.5 pct	MED
Time	Monsoon season drives highest revenue	MED

Table 21: Key Business Insights

10 Strategic Recommendations

No	Area	Recommendation
1	Retention	Win-back 85 Lost customers with personalised offers
2	Diversification	Invest in Furniture and Sports beyond Electronics
3	Digital Growth	Grow Online and Mobile App - 62.5 pct combined
4	Re-Engagement	Onboard New Customers within first 30 days
5	Loyalty	Reward 212 Weekly and Bi-Weekly buyers with perks
6	Expansion	Target Chennai 5.6 pct and Pune 8 pct campaigns
7	Seasonal	Amplify Monsoon season promotions - highest revenue
8	Payments	Offer UPI cashback and Credit Card EMI options

Table 22: Strategic Recommendations

11 Appendix - Pipeline Summary

Step	Name	Status	Notes
Step 0	Package Setup	COMPLETE	16 packages loaded
Step 1	Load Data	COMPLETE	9140 rows, 15 columns
Step 2	First Look	COMPLETE	15 columns verified
Step 3	Statistical Summary	COMPLETE	Skewness and distributions
Step 4	Missing and Duplicates	COMPLETE	0 missing, 0 duplicates
Step 5	Value Validation	COMPLETE	12 of 12 passed
Step 6	Data Cleaning	COMPLETE	9 actions applied
Step 7	Feature Engineering	COMPLETE	15 new features
Step 8	Customer Profiling and RFM	COMPLETE	300 profiles, RFM scored
Step 9	EDA Visualizations	COMPLETE	15 charts generated
Step 10	Save Outputs	COMPLETE	20 files saved

Table 23: Complete Pipeline Summary

Report generated automatically using R Markdown.
Raw data: `df_raw` | *Clean data:* `df` | *Profiles:* `customer_profile`